

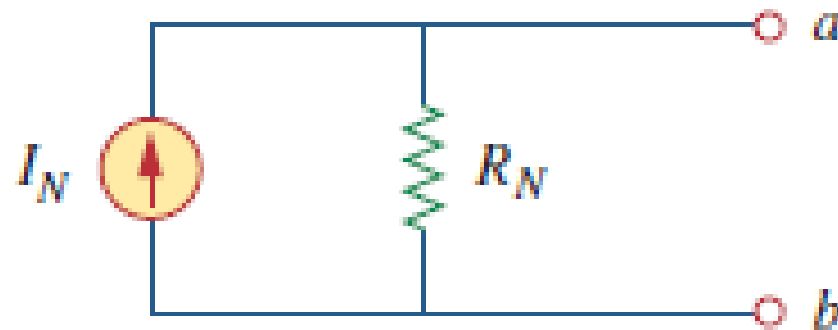
Module -1

DC Circuit Analysis and Theorems

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Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

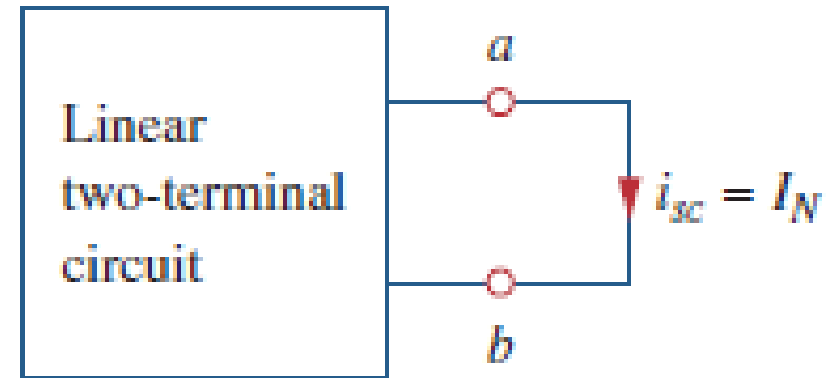


Norton's Theorem

$$I_N = i_{sc}$$

$$R_N = R_{Th}$$

$$I_N = \frac{V_{Th}}{R_{Th}}$$



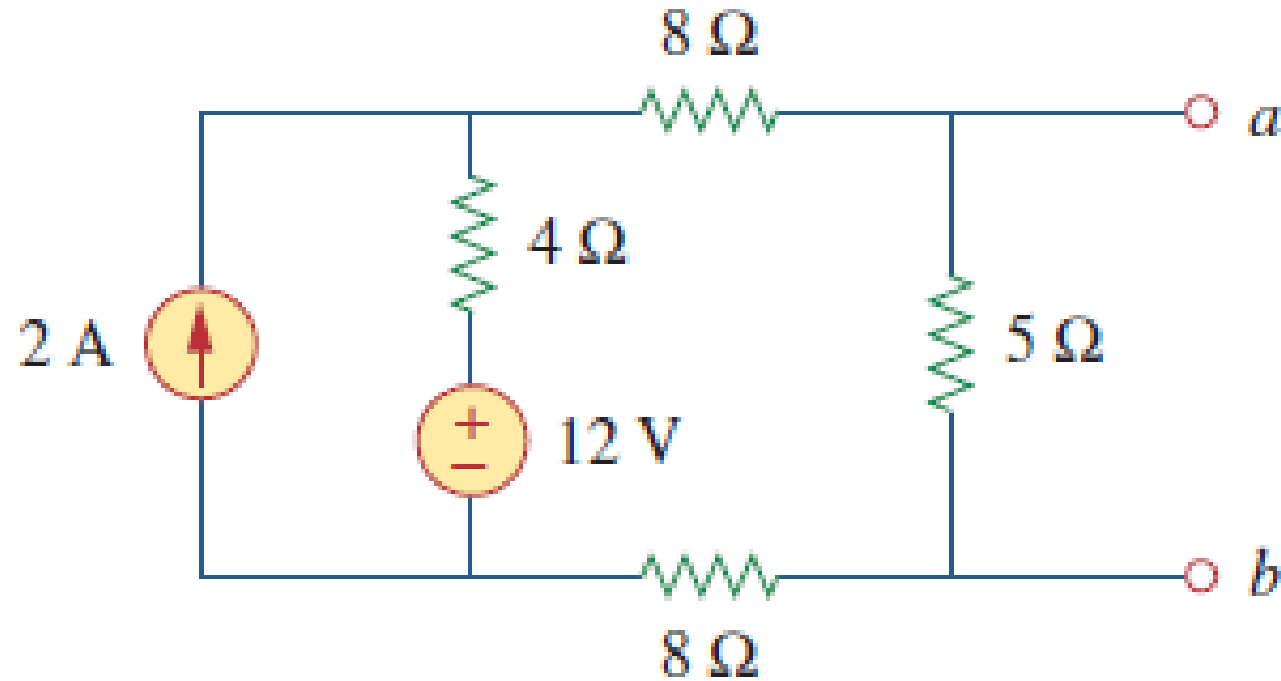
$$V_{Th} = v_{oc}$$

$$I_N = i_{sc}$$

$$R_{Th} = \frac{v_{oc}}{i_{sc}} = R_N$$

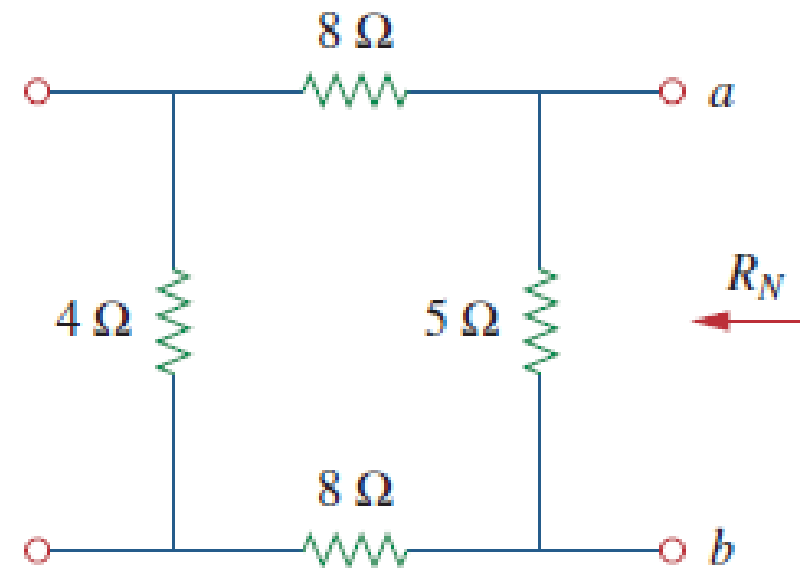
Tutorial - 1

Find the Norton equivalent circuit of the circuit at terminals a-b.



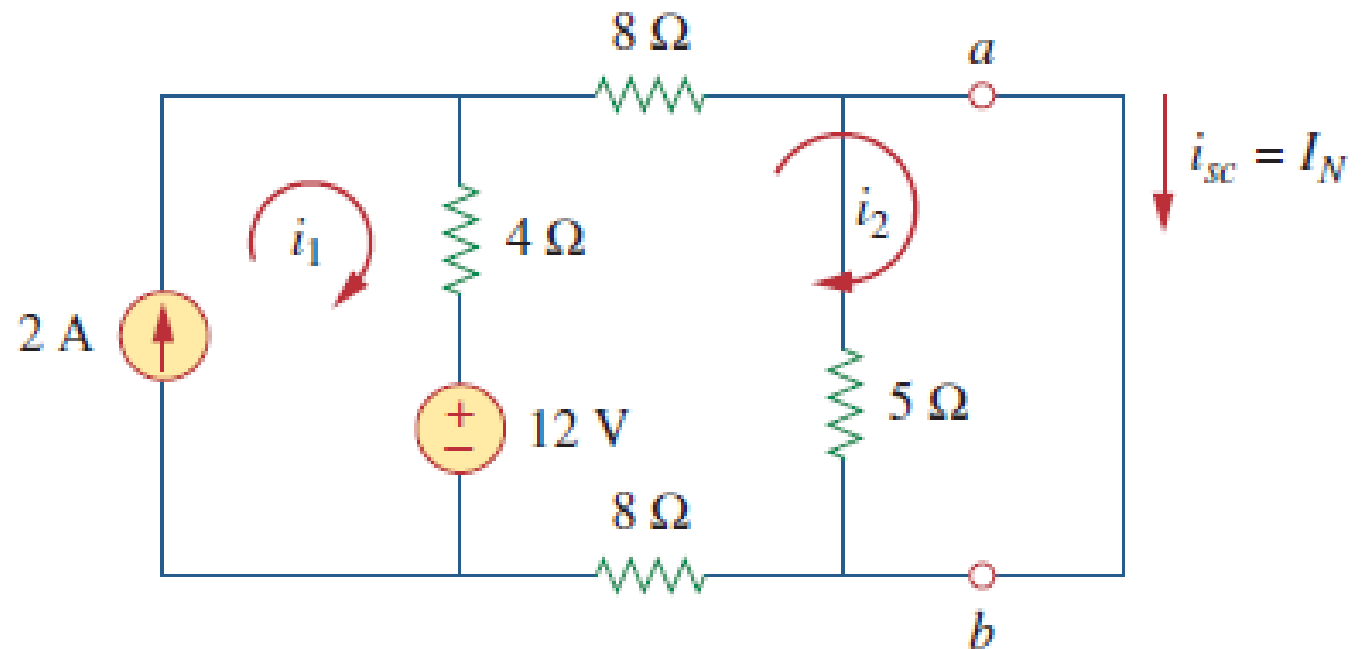
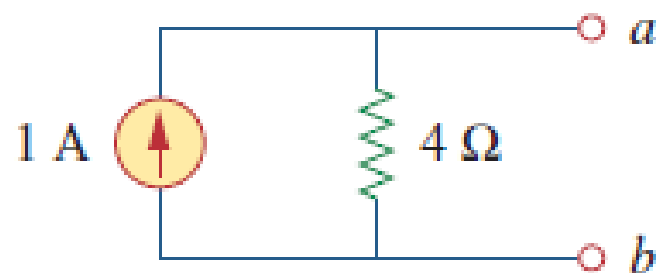
Solution:

$$R_N = 5 \parallel (8 + 4 + 8) = 5 \parallel 20 = \frac{20 \times 5}{25} = 4 \Omega$$



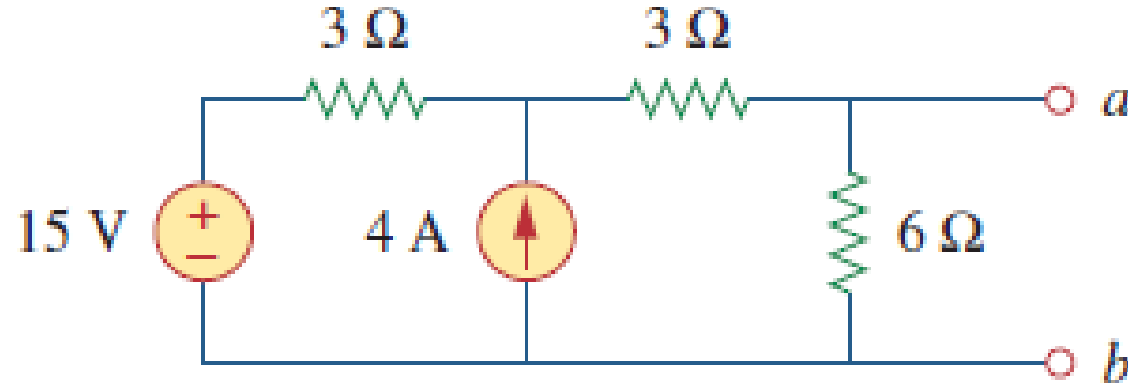
$$i_1 = 2 \text{ A}, \quad 20i_2 - 4i_1 - 12 = 0$$

$$i_2 = 1 \text{ A} = i_{sc} = I_N$$



Tutorial - 2

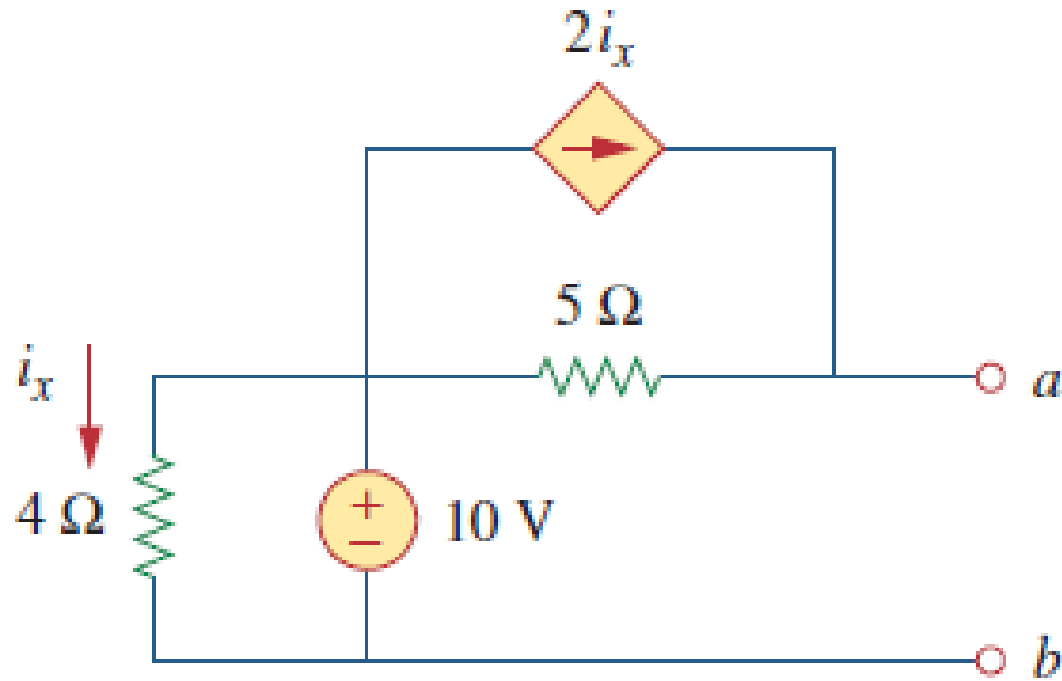
Find the Norton's equivalent of the circuit at the terminals a-b.



Answer: $R_N = 3\ \Omega$, $I_N = 4.5\ \text{A}$.

Tutorial - 3

Find the Norton's equivalent of the circuit at the terminals a-b.



Solution:

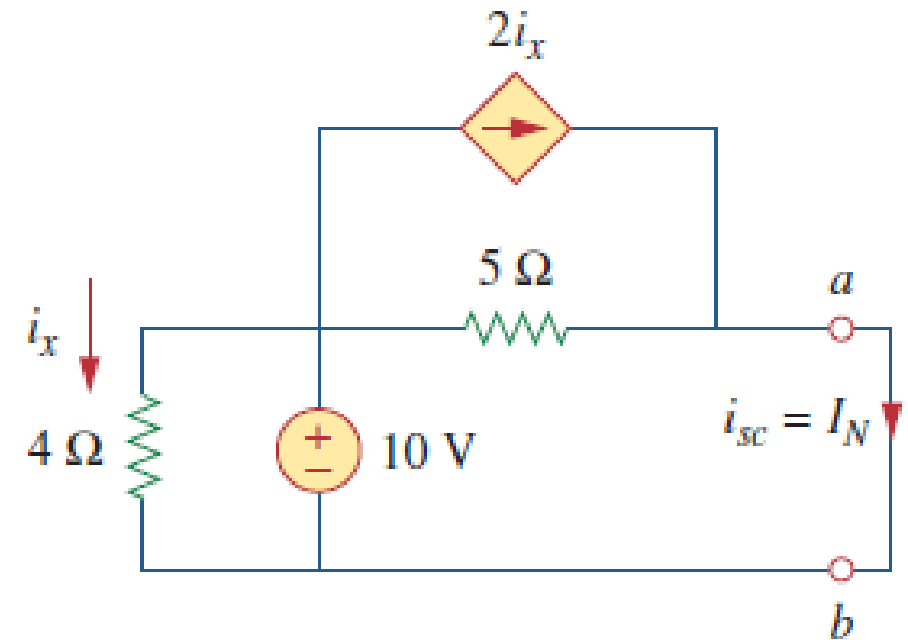
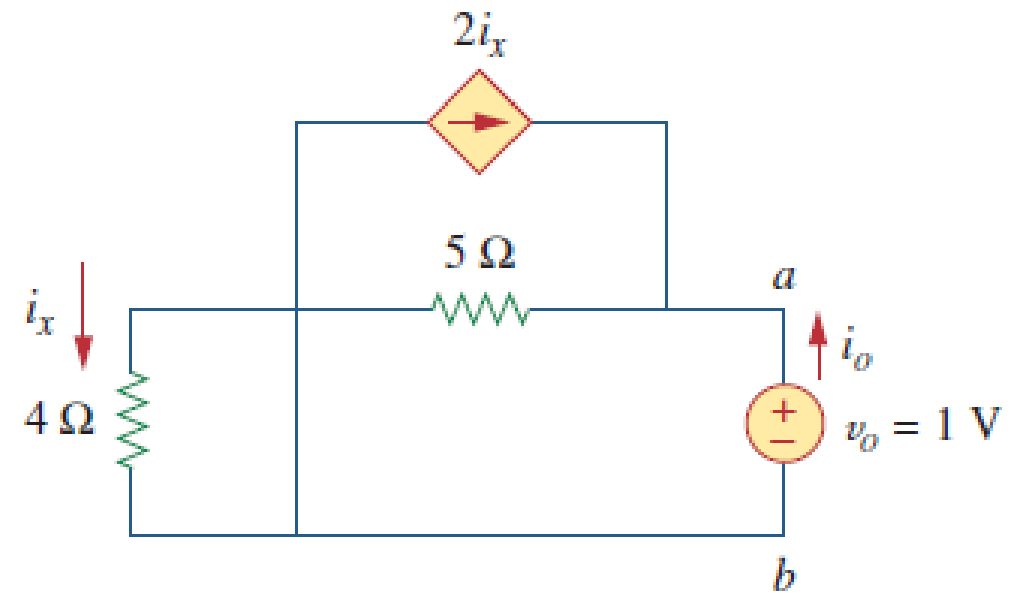
$$R_N = \frac{v_o}{i_o} = \frac{1}{0.2} = 5 \, \Omega$$

$$i_x = \frac{10}{4} = 2.5 \, \text{A}$$

At node a , KCL gives

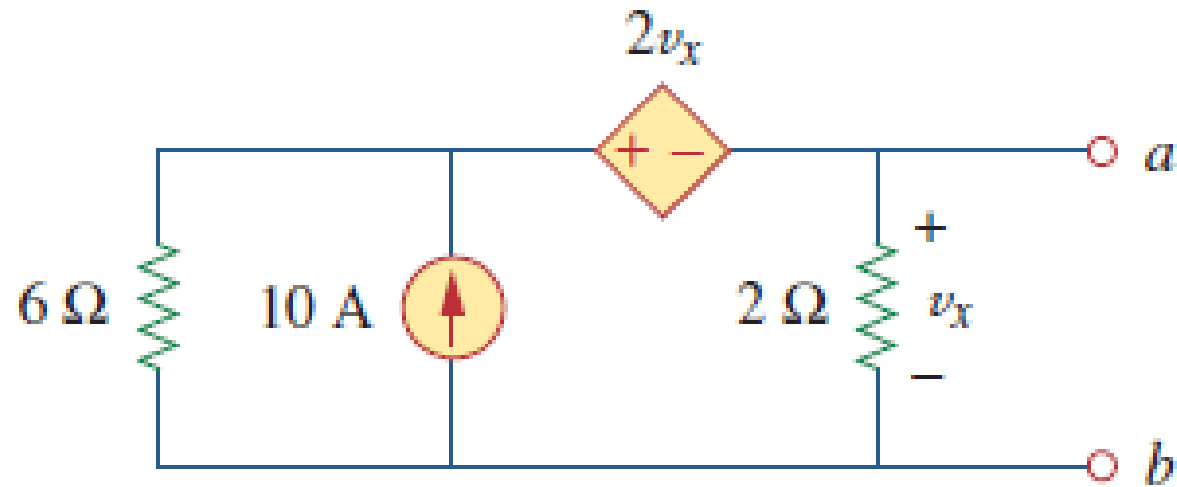
$$i_{sc} = \frac{10}{5} + 2i_x = 2 + 2(2.5) = 7 \, \text{A}$$

$$I_N = 7 \, \text{A}$$



Tutorial - 4

Find the Norton's equivalent of the circuit at the terminals a-b.



Answer: $R_N = 1\ \Omega$, $I_N = 10\text{ A}$.